CloudTrace Forensic Report

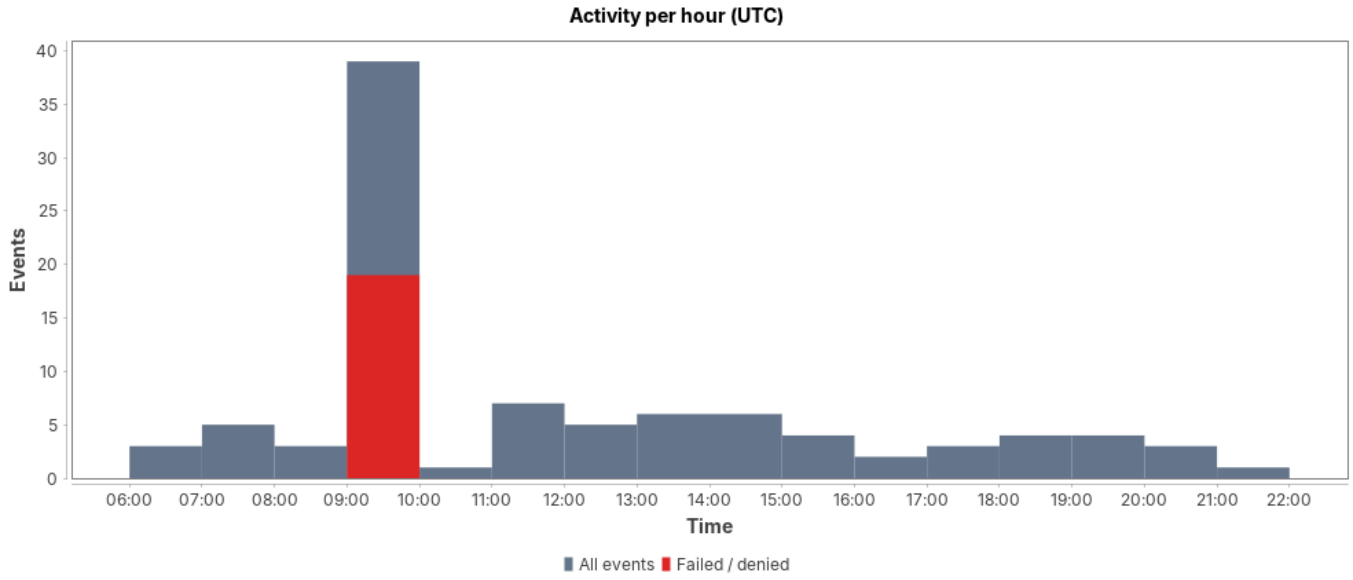
Case #1: Compromised CI user dev-jenkins

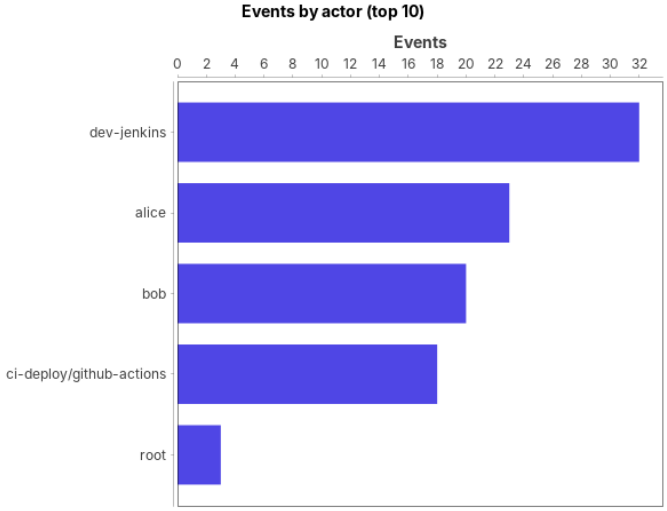
Investigator: anish Status: IN_PROGRESS Opened: 2026-11-10 18:08:26 UTC

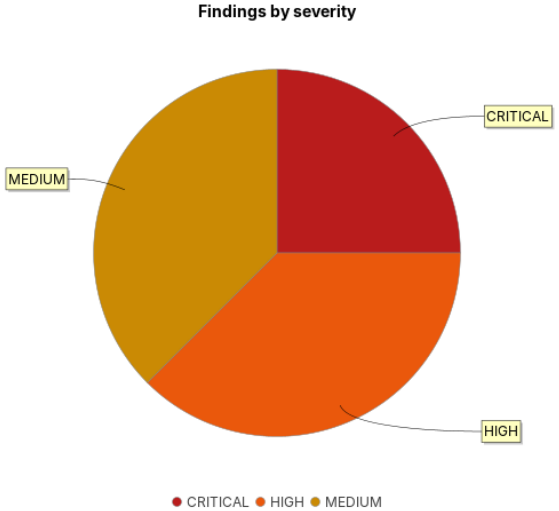
GuardDuty alert: console logins from a Tor exit node

Summary

Evidence records	96
Time span	2026-10-14 06:21:00 UTC to 2026-10-14 21:04:00 UTC
Distinct actors	5
Findings	8
Evidence integrity	VERIFIED (96 SHA-256 hashes match)







Findings

Severity	Rule	Actor	When	Details
CRITICAL	BRUTE-FORCE Repeated failed console logins	dev-jenkins	2026-10-14 09:12:00 UTC to 2026-10-14 09:18:00 UTC	7 failed console logins from 185.220.101.47 within 10 minutes, followed by a SUCCESSFUL login as dev-jenkins Evidence: 8 event(s)
CRITICAL	LOG-TAMPER Audit logging disabled or modified	dev-jenkins	2026-10-14 09:31:00 UTC to 2026-10-14 09:32:00 UTC	dev-jenkins performed 2 matching event(s): [StopLogging, DeleteFlowLogs] Evidence: 2 event(s)
HIGH	PRIV-ESC Possible privilege escalation	dev-jenkins	2026-10-14 09:26:00 UTC to 2026-10-14 09:28:00 UTC	dev-jenkins performed 3 matching event(s): [AttachUserPolicy, CreateAccessKey, CreateLoginProfile] Evidence: 3 event(s)
HIGH	SG-OPENED Security group opened to the internet	dev-jenkins	2026-10-14 09:34:00 UTC to 2026-10-14 09:34:00 UTC	dev-jenkins performed 1 matching event(s): [AuthorizeSecurityGroupIngress] Evidence: 1 event(s)
HIGH	ROOT-USAGE Root account used	root	2026-10-14 11:00:00 UTC to 2026-10-14 11:05:00 UTC	root performed 3 matching event(s): [ConsoleLogin, DeleteAccessKey, StartLogging] Evidence: 3 event(s)
MEDIUM	LOGIN-NO-MFA Console login without MFA	dev-jenkins	2026-10-14 09:18:00 UTC to 2026-10-14 09:18:00 UTC	dev-jenkins performed 1 matching event(s): [ConsoleLogin] Evidence: 1 event(s)
MEDIUM	RECON Burst of access-denied errors (reconnaissance)	dev-jenkins	2026-10-14 09:19:00 UTC to 2026-10-14 09:22:40 UTC	12 access-denied errors by dev-jenkins within 15 minutes across 6 different API calls Evidence: 12 event(s)
MEDIUM	S3-EXPOSURE S3 bucket access controls changed	dev-jenkins	2026-10-14 09:35:00 UTC to 2026-10-14 09:35:00 UTC	dev-jenkins performed 1 matching event(s): [PutBucketPolicy] Evidence: 1 event(s)

Chain of custody

Time	Action	By	Details
2026-11-10 18:08:26 UTC	CASE_OPENED	anish	Case created: Compromised CI user dev-jenkins
2026-11-10 18:08:27 UTC	EVIDENCE_COLLECTED	anish	96 new of 96 events from CloudTrail log files at sample/cloudtrail-incident.json, full time range (SHA-256 recorded per event)
2026-11-10 18:08:28 UTC	ANALYZED	anish	8 findings from 8 rules
2026-11-10 18:08:29 UTC	INTEGRITY_VERIFIED	anish	All 96 records match their hashes
2026-11-10 18:08:30 UTC	REPORT_GENERATED	anish	sample/sample-report.pdf sha256=ede844806f62a2f5f97abb0684ee734f220c600f6a383af356a7a702bf6e0596
2026-11-10 18:09:03 UTC	INTEGRITY_VERIFIED	anish	All 96 records match their hashes

Appendix: evidence log

Time	Event	Actor	Source IP	Result	SHA-256
2026-10-14 06:21:00 UTC	DescribeInstances	alice	203.0.113.10	OK	`4907b7a2c6a07dd57d24c652bc5be684ab5b0a53d031a26ff4a2d34cc43381a8`
2026-10-14 06:28:00 UTC	DescribeSecurityGroups	ci-deploy/github-actions	140.82.112.4	OK	`b1776a44399632320b26ff0fc8356fdb22a284470762e0172471c9d096ab55b2`
2026-10-14 06:40:00 UTC	DescribeInstances	bob	203.0.113.22	OK	`da22cbefd8d6a8019fbad746bc53a82c3df33b420128307218f75512492538d4`
2026-10-14 07:00:00 UTC	ConsoleLogin	alice	203.0.113.10	OK	`850d5657733e2371cba27a62e31893bf4a04cfe9f3aebda3e19aecab216bf6bd`
2026-10-14 07:01:00 UTC	ListBuckets	bob	203.0.113.22	OK	`a4aafb3c07c4195ca9be6cb3575a274e32390cb16df9bd0569cce35687245094`
2026-10-14 07:32:00 UTC	DescribeSecurityGroups	bob	203.0.113.22	OK	`b1d922ca170df1c3922fbc64bdcf25e02c984393f8a9dc4d22140218f17c896f`
2026-10-14 07:44:00 UTC	GetCallerIdentity	alice	203.0.113.10	OK	`07256b4336416f4768418372b49161064d5583205abfc15ba0c6c13321793f8a`
2026-10-14 07:46:00 UTC	DescribeSecurityGroups	ci-deploy/github-actions	140.82.112.4	OK	`ec340522dc3a5f75bfa1290989a1917c4729d8993e1254c283553e2d033d02d1`
2026-10-14 08:08:00 UTC	GetCallerIdentity	bob	203.0.113.22	OK	`6dc4eab861728f4e930dba6b820bf4012d039d91121c544d8271b6a5c0231dc0`
2026-10-14 08:12:00 UTC	ListBuckets	bob	203.0.113.22	OK	`61d0efca5b7dd9121b87ae6eb3282a736d72c0c6b01cab959b05b19d37430c65`
2026-10-14 08:30:00 UTC	GetCallerIdentity	bob	203.0.113.22	OK	`32d7160115e628ddda7304301d6387fc69a4ede17b74b55180cad234d14e681d`
2026-10-14 09:00:00 UTC	ListBuckets	alice	203.0.113.10	OK	`ef32fef25f8f7df833a636358b1d6d7eb9cc7c635eca124fe9cf40a0c2d0506a`
2026-10-14 09:12:00 UTC	DescribeInstances	ci-deploy/github-actions	140.82.112.4	OK	`ef550a510d6f4083fab6045e963ae984f9d7a269dc055afc72fee1898667bd83`
2026-10-14 09:12:00 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`df23ca2206df672130518ddce1c6dcc728405594f4546ca6e63d7f4b6648e8ca`
2026-10-14 09:12:40 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`517c62ed9848ba7f9282174fc45d7f0085fe5e179fc4a538d7333c8f3f27fe07`
2026-10-14 09:13:20 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`517464f041cb2157edf6d6aec733aec889498530f7bc39904c8fb54d13874f08`
2026-10-14 09:14:00 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`2ca5c99b7b028e14041ff6080f459a10b97211a408dd9d015ae871c504f2f8de`
2026-10-14 09:14:40 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`df4bd3e99aba07bdc3558a11ed4d011e56255b9fcb344686d402c848c4346ed7`
2026-10-14 09:15:20 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`14b570d67757625c0847cce774049c8717ee4cd4407bb8c1c2fc289dbf7f2018`
2026-10-14 09:16:00 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	Failed authentication	`ea33a0dc3ec33997925e0097a7c47f9ff38adc8372e0abbdf29842386b0e7622`
2026-10-14 09:18:00 UTC	ConsoleLogin	dev-jenkins	185.220.101.47	OK	`24534579b84f58cc79b7b07b6f85e54716a1ee82c56f392d6b89245d43f7ea05`
2026-10-14 09:19:00 UTC	ListUsers	dev-jenkins	185.220.101.47	AccessDenied	`8193886894dd2c24592af774ba9d6a3f0077d4925a98a2a208775bf9b4332289`
2026-10-14 09:19:20 UTC	ListRoles	dev-jenkins	185.220.101.47	AccessDenied	`fd4686c8e4ff8f277e426033ea84de36cda3b4556ee1473dce215015d82f76db`
2026-10-14 09:19:40 UTC	GetAccountAuthorizationDetails	dev-jenkins	185.220.101.47	AccessDenied	`683b8574fb1f35d7d069e101bf553599b8fd47aa9304edb9e90eece47c8180f8`
2026-10-14 09:20:00 UTC	ListSecrets	dev-jenkins	185.220.101.47	AccessDenied	`b7c85917de2e150cca8edb07a7427f667bd56cfc9fb7e72b88f7c28786c88cf0`
2026-10-14 09:20:20 UTC	DescribeDBInstances	dev-jenkins	185.220.101.47	AccessDenied	`f3eadb7b0c8833318ed3b052aa77c50931b997ba7cef7a43af4053c85b3c8897`
2026-10-14 09:20:40 UTC	ListKeys	dev-jenkins	185.220.101.47	AccessDenied	`08c40c64a0ab79d2b892c0aa6ead891a92ede9d9cd959783cce686e4b7b110a6`
2026-10-14 09:21:00 UTC	ListUsers	dev-jenkins	185.220.101.47	AccessDenied	`277482dcf67b57414250a3ea12ef3c6548b271b86717dfc6ed3633fca6286a0e`
2026-10-14 09:21:20 UTC	ListRoles	dev-jenkins	185.220.101.47	AccessDenied	`81a626fe848f18bec6e969655ffcc2261713e0778dee04f2371c63a74e29a3d7`
2026-10-14 09:21:40 UTC	GetAccountAuthorizationDetails	dev-jenkins	185.220.101.47	AccessDenied	`70bf58505c11ba5341f99789118a0e3cb7ff6e5665ac4e94f40b66d0416601dc`
2026-10-14 09:22:00 UTC	ListSecrets	dev-jenkins	185.220.101.47	AccessDenied	`5419b4d23ccaaaf297ddf4917f8fa176ca224279c3c91f8194ece0cc933a3d3a`
2026-10-14 09:22:20 UTC	DescribeDBInstances	dev-jenkins	185.220.101.47	AccessDenied	`43acfbb972971c9111cab9d504a7587626f853570095cf0c2c43d3f592cb580a`
2026-10-14 09:22:40 UTC	ListKeys	dev-jenkins	185.220.101.47	AccessDenied	`34861359ddd2ce890abcdf4bfd2e28793d123fce9f7b695fa9dd4cbc09ed82e1`

Time	Event	Actor	Source IP	Result	SHA-256
2026-10-14 09:26:00 UTC	AttachUserPolicy	dev-jenkins	185.220.101.47	OK	3c4f617338f3545b5f6e34f02ad1a1ca436d09f1ba1f391084c2cf6ddc1e36bf
2026-10-14 09:27:00 UTC	CreateAccessKey	dev-jenkins	185.220.101.47	OK	cfd48aa9c59f61840a4b7d6076238e0ac1213992e948a0b9fe2f9cbf1780c4af
2026-10-14 09:28:00 UTC	CreateLoginProfile	dev-jenkins	185.220.101.47	OK	fdd8be719c1519f8e98a501b4ba265c3badff7a0f5a770ac84df22ca0ea911b3
2026-10-14 09:31:00 UTC	StopLogging	dev-jenkins	185.220.101.47	OK	40ac426d2031ada60a81afd5838b1d42097f67933dc17709f523f2b428d9a794
2026-10-14 09:32:00 UTC	DeleteFlowLogs	dev-jenkins	185.220.101.47	OK	fa124a7ebc9236f232a53c9790118bed84b46e51f836bcdf15bf38b77e653151
2026-10-14 09:33:00 UTC	ListBuckets	ci-deploy/github-actions	140.82.112.4	OK	0e9f2a5a8b2469fed2b6501905282815d6d16c4663d42a1f661e3d7f85958460
2026-10-14 09:34:00 UTC	AuthorizeSecurityGroupIngress	dev-jenkins	185.220.101.47	OK	9b2416fa0857a3ed1cba6637c77f4a70d630c7c7f08325577f4a50e65fdeee15
2026-10-14 09:35:00 UTC	PutBucketPolicy	dev-jenkins	185.220.101.47	OK	8962f0874d3eab4b3d2e77072b0927f1edac89785f644c14c4158707fabe4b30
2026-10-14 09:37:00 UTC	GetObject	dev-jenkins	185.220.101.47	OK	932a2ea5afd9016ed6ac2c865c61a9ae3aefcf3e48530dc82f6da676e1df1720
2026-10-14 09:37:05 UTC	GetObject	dev-jenkins	185.220.101.47	OK	08980b19ad3e8b045cd7a711deed988179141cc9491a214a8a7627aa8de7f1b2
2026-10-14 09:37:10 UTC	GetObject	dev-jenkins	185.220.101.47	OK	36511b7b07eaf6abd58a939eea6e86d158e8f9a3a4cf6649d352fa656926e48f
2026-10-14 09:37:15 UTC	GetObject	dev-jenkins	185.220.101.47	OK	b5031c0cf38c83b155c703397a3f209a58e2a032ad0fa9602dfa6abd21de17f9
2026-10-14 09:37:20 UTC	GetObject	dev-jenkins	185.220.101.47	OK	b9fc1a69ec12aa77360a91547fff0702858a1468b9254d1a2a420db62397306b
2026-10-14 09:39:00 UTC	DescribeInstances	alice	203.0.113.10	OK	585f296428b6f1736dacd12df48cdbca6cc214ec72f2019d9c1273c8cbca6760
2026-10-14 09:45:00 UTC	ListBuckets	alice	203.0.113.10	OK	0bcee9cd656f49e292d41dfdd2c770a1dd6e38e55fe7d4db9425b73ff7bf0469
2026-10-14 09:48:00 UTC	GetCallerIdentity	bob	203.0.113.22	OK	af6f5ef89b1649fd99e9d1f6727ae7dc5bbfa67eb671ed40dbb673633c149743
2026-10-14 09:52:00 UTC	GetCallerIdentity	alice	203.0.113.10	OK	99cb9aa6d7b5f8f02583badc9af5f4f1488c5e26010460d0f80bdb11c46a6296
2026-10-14 10:56:00 UTC	ListBuckets	ci-deploy/github-actions	140.82.112.4	OK	a5f8bc68566d897196edf69988fa5c7d11d14f385d00274700492ee8c3837233
2026-10-14 11:00:00 UTC	ConsoleLogin	root	198.51.100.5	OK	f19ea9f619917947d3ae297b37051e71f9a7588f8004667388e3687aabb267cb
2026-10-14 11:03:00 UTC	DeleteAccessKey	root	198.51.100.5	OK	2593cd29929e0678d1b7b288d0373750b429d44c02bba0d47fcb9d962341141b
2026-10-14 11:05:00 UTC	DescribeInstances	bob	203.0.113.22	OK	2666285dbb388cc5cc410a04d14802ecaff8df03df5fa236a855e51766ce52fc
2026-10-14 11:05:00 UTC	StartLogging	root	198.51.100.5	OK	a83f9e08abb136deb98641c8dab72347a8cf42afa01e67ad4be556fa8d6ade12
2026-10-14 11:06:00 UTC	ListBuckets	alice	203.0.113.10	OK	4dced76355d3452a777449df88fed24215e0d45480991a4c4a48b2e7fb118722
2026-10-14 11:31:00 UTC	DescribeSecurityGroups	alice	203.0.113.10	OK	c8ecc91469e9b08d4f6233d8106e5fa736d0f564abfe85f3bf1fb8e04b21de71
2026-10-14 11:52:00 UTC	GetObject	ci-deploy/github-actions	140.82.112.4	OK	34fdf26e5aceee42bfab1504135dd7f1122dad672cf22199886b013caab70400
2026-10-14 12:12:00 UTC	DescribeInstances	bob	203.0.113.22	OK	1469b456c7f5559dbb32c1b219456f35775307707553f14986b2353c0ffb2266
2026-10-14 12:15:00 UTC	GetObject	alice	203.0.113.10	OK	619a69b5b8f19a871f7b477b16f4341ce3a4aa2f6a16b881e4aa239e4c658345
2026-10-14 12:25:00 UTC	GetObject	alice	203.0.113.10	OK	3f06ec339102668b6ab5efc0fcb0fd6fb1da51d9935bfcbcb4f77c9ef2ab0334
2026-10-14 12:37:00 UTC	ListBuckets	bob	203.0.113.22	OK	6ccd052635c56575d38a0eee32005c0b53dc9a393edbaf1b1bd2795435529d4b
2026-10-14 12:46:00 UTC	ListBuckets	alice	203.0.113.10	OK	23300c3e24fa2d968f0ff68a0a263f0c2990929acbab516f36daadd43acaa864
2026-10-14 13:05:00 UTC	DescribeSecurityGroups	ci-deploy/github-actions	140.82.112.4	OK	e1fd285efed51378bb8f4f60f5893096955605bc6cce5635ba28c7da82a5267b
2026-10-14 13:08:00 UTC	GetObject	alice	203.0.113.10	OK	0e05e332c9ab7bcb4a93563a92deee10ce0e35b69bfecfefa8877702dde88346
2026-10-14 13:09:00 UTC	GetObject	bob	203.0.113.22	OK	34e1ce494df9512f29c61e2063a10f64eb0dbbb97dc7c2603f1da472749b0e8a
2026-10-14 13:24:00 UTC	ListBuckets	alice	203.0.113.10	OK	df611d7ca77b1296aec70c5b8b61aacd9aafa0cb5f647d11799e0610be9314ed
2026-10-14 13:39:00 UTC	GetCallerIdentity	ci-deploy/github-actions	140.82.112.4	OK	71216732009f838ec1f046ef9d33be69dc1380f1d66e66b42382d6afc1344742
2026-10-14 13:57:00 UTC	DescribeSecurityGroups	ci-deploy/github-actions	140.82.112.4	OK	b58b5357cbb7b4e4d44d015377c05b0e3f52480640ef197ec7a71fb76274147d
2026-10-14 14:05:00 UTC	DescribeInstances	bob	203.0.113.22	OK	16bd46b144ae8de8492201a6f6be8b893be50196c9bf83fe33547407c3014001

Time	Event	Actor	Source IP	Result	SHA-256
2026-10-14 14:05:00 UTC	DescribeInstances	alice	203.0.113.10	OK	`710ea452f1e4a82045d2dcba897df5d62ea0eaa82df298285af62ce52ab74d2a`
2026-10-14 14:05:00 UTC	GetObject	bob	203.0.113.22	OK	`e4a246a2666c30c22c9c4bd2c1cc50e632dcd6d3c233c6efafc1fa39e4509922`
2026-10-14 14:14:00 UTC	GetCallerIdentity	bob	203.0.113.22	OK	`0a05dd42ceb3e8cfec25e507b74cd994009a16c163696a025555c9a0bdc908c4`
2026-10-14 14:28:00 UTC	ListBuckets	alice	203.0.113.10	OK	`0f369305a054edf8c802102948d2de57a8b294e5b94e4d5ec7596bca1c534f14`
2026-10-14 14:48:00 UTC	ListBuckets	alice	203.0.113.10	OK	`ed1230f0a9ebfb9f64470569e649ddd16661b830e360c7845ef312111ab5de7a`
2026-10-14 15:08:00 UTC	GetObject	alice	203.0.113.10	OK	`1bbab23e9e6cc1192686de9f5ea56857f65f4874bc567d64bbdbbc2bb1ee0d94`
2026-10-14 15:33:00 UTC	ListBuckets	bob	203.0.113.22	OK	`6754c6b40de6695272c1f4b32a7cb75128a9337c1b2da8126785d254cffa762f`
2026-10-14 15:48:00 UTC	GetCallerIdentity	ci-deploy/github-actions	140.82.112.4	OK	`8db383af8f33b0e92bebef10f3b3acf27a2d3520c9b1e3f17b1a0929f4889727`
2026-10-14 15:53:00 UTC	DescribeInstances	ci-deploy/github-actions	140.82.112.4	OK	`abd827a7a7a26708d8cbb58ca7059542818d5e8317c3e62726166d1bbd55b20d`
2026-10-14 16:04:00 UTC	ListBuckets	ci-deploy/github-actions	140.82.112.4	OK	`8c3efceb6758f100e8d8da9de7b7a31c3940a1926e7248b778f3e4ebeecb9cf0`
2026-10-14 16:25:00 UTC	DescribeSecurityGroups	alice	203.0.113.10	OK	`42bde6f12031344762d66b9cd57e0e8f58f56cd31a33ba0b85e65d697b9b6d76`
2026-10-14 17:02:00 UTC	DescribeSecurityGroups	bob	203.0.113.22	OK	`7448edd71dc6f36cc92883aa54a0c69df023b111b7ef7b5b9d4a27c927c6fb23`
2026-10-14 17:08:00 UTC	DescribeInstances	ci-deploy/github-actions	140.82.112.4	OK	`44ba83cafe4d19a34d0e188940fc971afbde2e1536646274b003cb761477aca7`
2026-10-14 17:24:00 UTC	DescribeInstances	ci-deploy/github-actions	140.82.112.4	OK	`a51fd15765a9629d7b7fd42127b1a9a07b601c124365a08c7415a4882b38607e`
2026-10-14 18:19:00 UTC	DescribeSecurityGroups	ci-deploy/github-actions	140.82.112.4	OK	`2f8f2a702fc5e9d77b4b3f51585e12f616a5a7235fa8ff72a917db6b9af63af7`
2026-10-14 18:22:00 UTC	ListBuckets	alice	203.0.113.10	OK	`d1dd77a32bf32624b5d8c1894718eaf1a97ed400d03524c6789bb6b96ceab4fd`
2026-10-14 18:37:00 UTC	DescribeSecurityGroups	alice	203.0.113.10	OK	`7ec3ba1751919e19d70c32a8b6a030bf975837634bc0768c6049ada9a16ed42c`
2026-10-14 18:47:00 UTC	GetObject	ci-deploy/github-actions	140.82.112.4	OK	`3933048ea1e51cbc2e1a8bf7f8d84ba2e9c634d842c7014d61d7bd9aa30f3cbc`
2026-10-14 19:15:00 UTC	DescribeSecurityGroups	ci-deploy/github-actions	140.82.112.4	OK	`95fd8f218126ff5559b75cf1ad072f6674f19bead9e6f206830474463ad85c2a`
2026-10-14 19:17:00 UTC	GetObject	bob	203.0.113.22	OK	`31b0e54908a0469882c9a72c31122ddfc5b35f2f251d66fcdc9ab5da852081f0`
2026-10-14 19:27:00 UTC	ListBuckets	alice	203.0.113.10	OK	`0fcb2cfa1648822aeff7e4f089f6f62039bff3cc28a6b209603782e99652084c`
2026-10-14 19:28:00 UTC	GetObject	bob	203.0.113.22	OK	`1f513b363a787cb3b2d896169f42795621f2d59c856f76dbafdb3665174de5ad`
2026-10-14 20:11:00 UTC	ListBuckets	bob	203.0.113.22	OK	`d67377aebf6b9280f5893dba1143e6b3ec2c5b5c8eb3aae81af00b93c9d9af0c`
2026-10-14 20:55:00 UTC	GetCallerIdentity	bob	203.0.113.22	OK	`fd9b588e09f0daf7cf1e5bf291672b9ec163d10968f8b87af43d4e022645c875`
2026-10-14 20:56:00 UTC	GetObject	ci-deploy/github-actions	140.82.112.4	OK	`49dc06d72b9e908506859443b33d60087d3c22890902d1f355cd5c6af09ca0a4`
2026-10-14 21:04:00 UTC	DescribeSecurityGroups	alice	203.0.113.10	OK	`c7af32a2d2b9c92e33d08c6b35dc461d846c32d132611fc267ee34b87bd8a074`